Math for CS & AI

Bobby

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Discrete Probability

Problem 1

Consider n aliens living on a planet with m days in a year. Let A be the event that at least two aliens share the same birthday. Prove that the collision probability $\Pr[A]$ satisfies:

$$\Pr[A] = 1 - e^{-N/m} + \mathcal{O}\left(\frac{1}{\sqrt{m}}\right)$$

as $m \to \infty$, where $N := n(n-1)/2$.

Hint 1

It is much easier to first calculate the probability of no collisions, denoted as $\bar{A}$.

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Hint 2

The asymptotic behavior of the error term depends heavily on the growth rate of n. Decompose the proof into two cases based on a threshold:

- Case 1: $n \leq m^{0.6}$ (Small n)
- Case 2: $n > m^{0.6}$ (Large n)

Hint 3

Take the natural logarithm of $\Pr[\bar{A}]$ and apply the Taylor expansion $\ln(1-x) = -x + \mathcal{O}(x^2)$:

$$\ln \Pr[\bar{A}] = \sum_{k=0}^{n-1} \ln\left(1 - \frac{k}{m}\right) = \underbrace{-\sum_{k=0}^{n-1} \frac{k}{m}}_{\text{Main term } -N/m} + \underbrace{\mathcal{O}\left(\sum_{k=0}^{n-1} \frac{k^2}{m^2}\right)}_{\text{Error term } \mathcal{O}(n^3/m^2)}$$

Hint 4

Use the standard inequality $1 - x \leq e^{-x}$ for all $x \in \mathbb{R}$ to upper bound the product directly:

$$\Pr[\bar{A}] \leq \prod_{k=0}^{n-1} e^{-k/m} = e^{-N/m}$$

Then, show that for large n, both $\Pr[\bar{A}]$ and $e^{-N/m}$ become negligibly small and are absorbed by the $\mathcal{O}\left(\frac{1}{\sqrt{m}}\right)$ error bound.